MangroveGuard: A Deep Learning Approach to *Mangrove* Stability Assessment Utilizing YOLOv8-Nano

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By:

Ivan Kly B. Lamason

Elzen Rein Marco Maceda

Dan Coby G. Tabao

Vincent N. Pensader

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CHAPTER I

This chapter will discuss the introduction of the study. It provides a comprehensive overview of the study outlining its objectives, purpose, description, scope and limitation.

Introduction

Spotted Mangrove, also known as *Rhizophora Stylosa*, are one of the species of mangroves that is commonly planted near coastal areas in Leyte. Mangroves trees grows in the coastline of tropical and sub-tropical countries and is known for its adaptability to extreme conditions such as high salinity and temperature (Goloran, Demetillo, & Betco, 2020). It also provides shelter to aquatic creatures, since most of the trees are planted near coastal areas. Fish, crustaceans and other species live in roots and between tree trunks (Cudiamat & Valdez, 2022). It reduces waves and storm surges during typhoons which serves as a first line of defense against flooding and erosion (Menendez, J, Saul, Narayan, & Beck, 2020). Without the help of the trees, many people, especially living near coastal areas, are likely to be affected by such disasters. They serve as critical biological frontiers, providing essential services to the people. These dense forests of mangroves are also considered as high-priority ecosystems for a number of recent large international conservation initiatives (Friess, et al., 2020).

Despite the ecological significance, there are many factors that can harm such forests. Other than storm surges and natural disasters, climate change and human disturbances also affect the trees integrity and stability. These disturbances are usually caused by economic, socio-cultural, and technological changes in society (Anwar, Setyasih, Ardiansyah, Dwi, & Wibowo, 2021). Identifying the stability of the mangroves on manual assessment of such problems. Locals, environmental scientist, or

field workers does an on-site inspection where the mangrove is planted, gather the necessary data to assess the stability of the tree.

Machine learning has been widely explored for ecological monitoring. It faced significant limitations in terms of real-time, on-device assessment, fine-scale structural measurement and mobile deployment in remote environments. Most ecological monitoring systems often rely on cloud-based servers, high-end computational devices, or large remote sensing platforms such as satellites and drones. The model requires substantial GPU and training time demands, which is impractical in remote mangrove swamps and can contribute to environmental degradation (Liu, et al., 2024).

In recent years, the integration of spatial computing and deep learning has revolutionized environmental monitoring. These algorithms have the ability to analyze vast quantities of data, decipher complex relationships, and predict future events, which offers promising capabilities to implement technologies based on more precise and reliable data processing (Binetti, Massarelli, & Uricchio, 2024). International research highlights the efficacy of Computer Vision (CV) in automating plant phenotype analysis. It extracts useful information from plants images and videos, which has become an essential technique in plant phenomics (Mochida, et al., 2019). Yet many existing models are too computationally expensive for standard mobile devices or rely heavily on stable cloud connectivity, which is non-existent in dense mangrove swamps. It requires high bandwidth and low latency for wireless communication, which is imposed as one of the challenges to design, configuration and optimization (Wu, Li, & Deng, 2020). Innovations in "edge AI," such as the YOLO (You Only Look Once) architecture and quantized neural networks, present an opportunity to bring high-precision ecological assessment directly to the field. Accurate and efficient object

detectors on edge-devices are essential for real-time space application (Altaie, Abdelkareem, & Alhasanat, 2025).

This study introduces MangroveGuard, a mobile edge-computing solution designed to classify the structural stability of Rhizophora stylosa through a quantized YOLOv8-Nano model. By leveraging Flutter and LiteRT, the application performs on-device instance segmentation to extract morphological root features, calculating a weighted stability score in real-time. This framework provides stakeholders in the Philippines with an application that helps local conservation efforts with a low-cost method for assessing complex mangrove architectures without the need for manual physical measurementsopera

1.2 Project Purpose & Description

The mangrove forests in our country serves as a first line of defense during storm surge or typhoons. Knowing their stability and integrity will help us be aware of how long the trees will hold on after many disasters. Many individuals will be affected if these forests are not in their stable conditions to continue being the biological frontiers of the community. The study will be designed to address such issues. It checks the tree itself based on the data gathered about the mangrove trees that makes them vulnerable.

The study will develop MangroveGuard, a seamless mobile solution designed demonstrates a system that uses edge-AI instance segmentation to analyze still images on-device, assessing mangrove tree stability by extracting structural indicators (Root Count, Root Density, Root Spread, Root Coverage Ratio, Structural Symmetry, and Root Thickness Proxy) and computing a weighted Stability Score. This automated process serves as the primary basis for determining a quantified Stability Score, which

categorizes the tree's structural strength against environmental stressors, such as storm surges and coastal erosion without the need for invasive physical measurements or cloud-based processing. The younger trees or the saplings will be exposed to waves, meanwhile, older trees are taller and are exposed to winds (Gijón Mancheño, et al., 2024). Root production was greatest for older mangroves, where live roots reach depths of 1.0 m in forested zones than typically studied. (Conroy, Kelleway, & Rogers, 2025). Soils near mangrove forest are typically soft, waterlogged, and weak, making anchorage difficult. The root system of the mangrove provides a network of physical reinforcement, which stabilizes the tree and the surrounding soil (Karimi, et al., 2022). It will involve locals and environmental scientist as participants to test the application's usability and accuracy. The investigation will be conducted in selected mangrove protected areas in Leyte, Philippines, providing a real-world testing ground characterized by complex lighting and dense vegetation of trees. By aligning spatial computing with ecological monitoring, this project addresses the technical gap in high-accessibility, on-device tools for field-based environmental assessment.

1.3 Objective of the Study

The aim is to develop and evaluate a mobile-based system titled **MangroveGuard** that utilizes the YOLOv8-Nano deep learning model to classify and assess mangrove stability by quantifying aerial root density using standard mobile devices. The study also aims to:

1. Collect a comprehensive dataset of Rhizophora Stylosa images across protected areas in Leyte to serve as the ground-truth for MangroveGuard. Captured via smartphones under diverse field conditions, each image is annotated with

mangrove_trunk and mangrove_root classes to document varying root complexities.

2. Train a YOLOv8-Nano instance segmentation model to achieve an accuracy of at least 80-90% for identifying and classifying mangrove trees. Use standard evaluation metrics including mAP, precision, recall, F1-Score, and a confusion matrix to verify results, ensuring the architecture remains lightweight enough for real-time performance on mid-range Android devices.
3. Evaluate the 'MangroveGuard' application using ISO 9241-11 usability standards for field researchers in terms of the following:
   3.1. Efficiency
   3.2. Effectiveness
   3.3. Satisfaction

4. Conduct Black-box testing for functional requirements system outputs and White-box testing to verify internal logic, code structure and model integrity

1.4 Scope and Limitation

The study focuses on the development of a mobile-driven platform titled “MangroveGuard”. It is designed to assess the stability of mangrove trees through the analysis of visible structural characteristics. The application utilizes deep learning to support non-invasive and accessible monitoring of the mangrove ecosystem. MangroveGuard evaluates structural stability by analyzing the spatial relationship between root spread and the trunk centroid. By quantifying the distribution and density of mangrove root instances relative to the mangrove trunk, the system generates a stability score, which is categorized into Low Stability, Moderate Stability, and High Stability levels. This multi-factored approach, incorporating structural symmetry and

root thickness proxy, provides a high-precision indicator of structural resilience, a critical factor in determining a mangrove's ability to withstand environmental stressors and mechanical damage. The investigation will be conducted in selected mangrove protected areas in Leyte, Philippines, specifically covering the municipalities of Tacloban and Palo. These sites provide a real-world testing ground characterized by complex lighting, dense vegetation of trees, and natural environmental obstruction. The application operates entirely on mobile devices, enabling real-time assessment during fieldwork in remote mangrove areas where internet connectivity is limited or inaccessible. The study is calibrated for mangrove environments and is limited to the evaluation of above-ground features observable through standard mobile cameras.

The study is limited to the analysis of visible, above-ground morphological features captured via standard 2D mobile photography, which introduces the constraint of spatial occlusion where roots positioned directly behind the trunk or obscured by dense overlapping structures remain undetected and excluded from Root Count and Root Spread calculations. Furthermore, the system does not account for the surface texture, internal wood density, or subsurface factors such as underground root architecture, soil composition, and sediment stability, which are critical biomechanical determinants of stability. Environmental variables inherent to mangrove ecosystems including uneven lighting, tidal interference, and varying mobile camera resolutions may further influence the consistency of the YOLOv8-Nano detection performance. Consequently, the resulting Stability Score serves as a non-invasive image-based proxy rather than a definitive biomechanical measurement, positioning the application as a preliminary, high-accessibility tool intended to complement, rather than replace, comprehensive evaluations conducted by trained environmental scientists and forestry professionals.

CHAPTER II

The chapter reviews existing research relevant to the project, a mobile application designed to assess mangrove stability. This review aims to provide a clear foundation for our work, identifying current research gaps and explaining how MangroveGuard addresses them.

Review of Related Literature and Systems

2.1 Related Literature

2.1.1 Ecological Role of Mangroves as Coastal Protection and Habitat Support

Mangroves are vital coastal ecosystems that provide essential services such as shoreline stabilization, storm surge attenuation, erosion control, carbon sequestration, and habitat provision for diverse marine and terrestrial species. Recent reviews highlight that mangrove act as natural barriers against extreme weather events and rising sea levels, effectively reducing coastal vulnerability and protecting human settlements (K et al., 2024; Akram et al., 2023; Dabalà et al., 2023). For example, prioritizing mangrove conservation can safeguard billions in property value and protect millions of people from coastal hazards (Dabalà et al., 2023). Additionally, mangroves serve as nurseries for fish and invertebrates, support biodiversity, and contribute to blue carbon storage critical for climate mitigation (K et al., 2024; Akram et al., 2023; Wang & Gu, 2021). However, these benefits are threatened by anthropogenic pressures such as urbanization, aquaculture expansion, and pollution (Akram et al., 2023; Ahmed et al., 2022; Wang & Gu, 2021). The degradation of mangroves directly reduces their capacity to buffer coastlines and maintain ecosystem services, which underscores the importance of assessing mangrove stability, as unstable stands diminish coastal protection functions (K et al., 2024; Akram et al., 2023; Dabalà et al., 2023).

2.1.2 Structural Stability Indicators in Mangroves

The structural stability of mangrove trees is primarily determined by their root systems, specifically metrics such as root count, density, spread, coverage ratio, structural symmetry, and thickness. Recent studies emphasize that dense, well-distributed roots are essential for anchoring trees against windthrow and wave action, thereby enhancing resistance to storms and erosion (Ahmed et al., 2022; Wang & Gu, 2021). Research has shown that salinity stress can reduce stand density, tree height, and overall productivity, demonstrating how abiotic factors influence the structural attributes linked to stability (Ahmed et al., 2022). Furthermore, root architecture, including spatial arrangement and density, directly affects a tree's ability to withstand hydrodynamic forces in intertidal zones (Wang & Gu, 2021). Landscape-scale analyses also show that patch density and shape indices can be used to assess disturbance levels in mangrove stands, effectively linking spatial structure with ecosystem health (Toosi et al., 2022).

2.1.3 Computer Vision in Plant Phenotyping and Ecological Monitoring

Advances in computer vision have revolutionized plant trait analysis by enabling the automated extraction of morphological features from images. Deep learning models, including convolutional neural networks (CNNs) and semantic segmentation architectures, can accurately delineate plant organs such as roots and trunks from complex backgrounds (Maurya et al., 2021; Wang et al., 2024; Schürholz et al., 2023; Lassalle & De Souza Filho, 2022). For instance, CNNs applied to aerial imagery can segment individual mangrove tree crowns with high precision, allowing for the estimation of canopy height, which is a key indicator of biomass and health (Schürholz et al., 2023). Additionally, lightweight segmentation networks like

UrmsNet have been introduced for efficient species-level identification using RGB images while maintaining high accuracy at a low computational cost (Wang et al., 2024). These methods enable the automated measurement of spatial features—such as centroid location for symmetry and area calculation for coverage ratio—directly from field images without manual intervention (Maurya et al., 2021; Schürholz et al., 2023; Wang et al., 2024).

2.1.4 Edge-AI Feasibility for Remote Ecosystem Assessment

Deploying AI models on mobile devices ("edge AI") allows for real-time analysis without reliance on cloud connectivity, which is a crucial advantage in remote environments like mangrove forests. Recent research highlights the feasibility of running deep learning models offline on smartphones or drones for ecological monitoring tasks (Wang et al., 2024; Bakar et al., 2024; De Souza et al., 2024). Specifically, lightweight segmentation networks can achieve high accuracy with low computational requirements suitable for mobile deployment (Wang et al., 2024). Edge-AI approaches address common fieldwork challenges such as intermittent internet access or power constraints and enhance data privacy by processing sensitive environmental data locally (De Souza et al., 2024; Bakar et al., 2024). This justifies the design of systems that operate fully offline on mobile devices to enable rapid assessment directly at remote sites.

2.2 Related Systems

2.2.1 On-Device Computer Vision Systems using Lightweight Inference Runtimes

Recent systems emphasize the deployment of optimized models like YOLO for real-time tasks on resource-constrained hardware. Edge YOLO is designed for real-time object detection in autonomous vehicles by optimizing model size and

computation for low-latency inference on edge devices (Wang et al., 2025). Similarly, evaluations of YOLOv8-Nano on platforms such as NVIDIA Jetson and Raspberry Pi demonstrate that these models can achieve up to 65 FPS through quantization, highlighting the trade-offs between speed and energy consumption (Rey et al., 2025). The Lightning Rod agent facilitates local data collection and incremental inference within IoT networks, although it is general-purpose and lacks specialized ecological feature extraction (Ficili et al., 2025). Furthermore, Tiny AI models have shown the feasibility of deploying advanced vision models for medical diagnosis across heterogeneous mobile hardware, demonstrating a design philosophy aligned with this study (Rancea et al., 2024).

GVC-YOLO (Zhang et al., 2024) is a lightweight real-time detector optimized from YOLOv8n for edge computing to detect cotton aphid damage. The model incorporates GSConv and VoVGSCSP modules along with coordinate attention and SimSPPF to preserve detection accuracy while significantly reducing model size and FLOPs. Despite its efficiency, the system is evaluated primarily on Jetson Xavier NX hardware and focuses on a specific agricultural problem, limiting its validation for smartphone-class devices and cross-domain applications. Nevertheless, its techniques for model compression and optimization are directly relevant to MangroveGuard, particularly in developing a compact YOLOv8-Nano variant for on-device mangrove trunk and root monitoring.

The Fruit Health Detection using YOLOv8 system (Patil et al., 2025) combines YOLOv8 for rapid fruit detection with downstream convolutional neural networks for classification of fruit health. The system is designed for continuous monitoring in agricultural environments and emphasizes efficient edge deployment. However, it does not provide detailed benchmarks for low-power smartphones and does not include

instance segmentation of plant structures, making its applicability to trunk and root analysis uncertain. Its relevance lies in demonstrating an edge-focused pipeline and the effective use of transfer learning, which can inform the training of compact models for mangrove monitoring tasks.

The efficient palm oil tree detection and counting system (Farhan & Akhtar, 2025) applies YOLOv8 to detect and count trees with a focus on achieving real-time performance on edge devices. The study validates its approach by comparing accuracy and speed against other YOLO-based methods. However, it does not explore segmentation of trunks or roots and lacks integration into a mobile application. Despite these limitations, it addresses a similar domain of tree monitoring, offering valuable insights into model design for efficient detection and counting of mangrove stems in field environments.

A Mobile-Based Deep Learning Approach for Mango Leaf Disease Detection (Bhat & Swetha, 2025) demonstrates the deployment of a convolutional neural network using TensorFlow Lite within a Flutter mobile application. The model uses quantization-aware training to reduce size and improve inference efficiency while maintaining accuracy across a dataset of 4,962 images. However, the system is limited to classification of leaf diseases and does not perform detection or segmentation of structural components. Its primary relevance lies in presenting a complete mobile deployment pipeline, including offline inference, which is directly applicable to the development of MangroveGuard.

2.2.2 Instance Segmentation Pipelines for Plant/Root Feature Extraction

Sophisticated segmentation architectures are increasingly used for detailed plant analysis, though they often require high computational power. PSegNet (Li et al., 2022)

performs simultaneous semantic and instance segmentation on 3D plant point clouds, enabling precise structural analysis. However, it relies on LiDAR data, which is not practical for standard smartphone-based applications. MaskFormer (Cheng et al., 2021) introduces a unified approach to semantic and instance segmentation through mask classification, achieving high accuracy across various benchmarks. Despite its effectiveness, its computational requirements typically necessitate server-side deployment, making it unsuitable for lightweight mobile environments.

For mangrove-specific applications, architectures such as Capsules-UNet and GC-UNet (Lin et al., 2025) enhance spatial feature representation and capture long-range dependencies in multispectral imagery. However, these models are designed for large-scale mapping rather than close-range trunk or root analysis. Similarly, UAV-based LiDAR systems (You et al., 2023) can segment individual trees and extract structural parameters but depend on specialized sensors and struggle in dense environments.

Across these studies, there is insufficient evidence of dedicated instance segmentation systems targeting plant roots or trunks using standard RGB images within mobile constraints. Most works focus on detection or classification tasks rather than detailed structural segmentation. This limitation highlights a significant research gap.

For MangroveGuard, this gap implies the need to adapt detection-based approaches such as YOLOv8-Nano or integrate lightweight segmentation techniques while developing custom datasets for trunk and root structures. Addressing this limitation represents a key innovation of the proposed system.

2.2.3 Mobile UI Development with Flutter

The integration of Flutter with TensorFlow Lite has enabled the development of cross-platform mobile applications capable of running AI models locally. These frameworks support real-time camera input and allow compressed models to perform inference without cloud connectivity (Wang et al., 2025; Rey et al., 2025). Flutter also enables visualization of detection outputs such as bounding boxes, facilitating intuitive user interaction in field-based applications (Gill et al., 2024).

The WAISMAIS IoT-Enhanced Corn Disease Diagnosis system (Baste et al., 2025) integrates Flutter with IoT-based soil nutrient analysis and YOLOv8 for disease detection. While it demonstrates strong multi-platform capabilities, the study notes that offline inference remains a future enhancement, indicating that full edge deployment was not achieved. This highlights the importance of designing fully offline-capable systems like MangroveGuard.

A real-time banana harvest readiness prediction system (Baglat et al., n.d.) utilizes a squeeze-and-excitation enhanced YOLO model within a Flutter application. The system enables real-time predictions in the field, though detailed performance metrics are not provided. Its relevance lies in demonstrating end-to-end integration of YOLO-based models within a mobile interface.

Plant Guard: AI-Enhanced Plant Diseases Detection (Arulmurugan et al., 2024) uses MobileNet, TensorFlow, and OpenCV in a Flutter application to achieve high accuracy in disease detection. However, it focuses on classification rather than structural analysis. Similarly, Cotton Guard (Arthy et al., 2024) proposes a Flutter-based system with both online and offline modes, though it lacks implementation results. The Mango Leaf Disease Detection using YOLOv5 Mobile App (Renugadevi

et al., n.d.) integrates a YOLO-based detector into a Flutter application for rapid field assessment. While it lacks detailed benchmarking, it confirms the feasibility of deploying YOLO-family models on mobile device

CHAPTER III

This chapter provides a detailed discussion of the technical components involved in the study. It outlines the current manual processes and high-end technological dependencies, while introducing the proposed "MangroveGuard" system, including the specific software, hardware, and people-ware required for its implementation

TECHNICAL BACKGROUND

3.1 Current System

The current methods for assessing mangrove stability in selected protected areas of Leyte primarily rely on manual field observations and high-cost remote sensing. Based on the Department of Environment and Natural Resources (DENR) of Region VII, they assess the mangrove areas using the Belt Transect Design. Environmental scientists and coastal rangers conduct on-site inspections involving physical measurements of tree diameter, height, and visual estimations and distances. Other than manual measurements, they also compute the stability manually based on the data they have gathered in the forests. Their current limitations on assessing the stability are the inclement on weather conditions and tidal influence. These manual processes are time-consuming, prone to human subjectivity, and difficult to standardize across different observers. While advanced alternatives such as satellite imagery (Remote Sensing), LIDAR-equipped drones, and high-performance GPU-based deep learning models exist, they present significant barriers for local stakeholders and to the environmental scientist. These systems often require stable high-speed internet for cloud processing, expensive computational hardware for real-time inference, and specialized technical

expertise to interpret complex data. In remote mangrove swamps where connectivity is non-existent and mobile portability is a necessity, these high-end technological solutions are often impractical for immediate, ground-level assessment.

3.2 Proposed System

The proposed system, MangroveGuard, is a comprehensive mobile tool designed to assist environmental researchers and locals in monitoring the resilience of a mangrove tree (Rhizophora stylosa). Developed using Flutter, the application provides a seamless and responsive user experience across both Android and iOS devices, ensuring that field workers have a reliable tool regardless of their preferred mobile platform. The system is engineered to operate entirely offline, acknowledging that mangrove habitats are often located in remote coastal areas with little to no cellular connectivity. By processing all data directly on the user's device, the system guarantees immediate feedback and data privacy without relying on cloud servers.

At the heart of the system is an advanced computer vision engine powered by TensorFlow Lite. This technology allows the application to function like a "smart eye"; when a user points their camera at a tree, the system uses a sophisticated Artificial Intelligence model to instantly recognize and separate the tree's trunk from its complex root system. This process, known as instance segmentation, happens in real-time on the phone. It converts the image into data, mapping out the precise geometry of the tree's structure without needing to send images to the internet for processing.

To assess the structural integrity of the mangrove, the system employs a custom Stability Score (S) algorithm. Once the AI has mapped the tree, the system analyzes six critical structural indicators: the Root Count (RC), the Root Density (RD), the Root

Spread (RS), the Root Coverage Ratio (RCR), the Structural Symmetry (SS), and the Root Thickness Proxy (RT). By combining these factors, the system calculates a single stability rating categorized as Low Stability, Moderate Stability, or High Stability. This provides researchers with an objective, standardized metric to compare trees over time, replacing subjective visual estimates with concrete, calculated data. Finally, the system includes robust features for data management and reporting. All scan results are saved locally to the device's secure internal storage, allowing users to build a historical log of their field surveys and view trends through interactive charts. To facilitate collaboration, the app includes a built-in reporting engine that compiles the visual data, the AI-generated overlay, and the calculated stability metrics into a professional PDF document. This allows researchers to instantly generate and share detailed field reports with colleagues or stakeholders directly from the site.

3.2.1 Software

The following table outlines the software tools and frameworks essential for the development and operation of the MangroveGuard system.

Table 1: *Software Requirements*

Software	Features	Purpose
Flutter SDK	Cross-platform UI development, Hot Reload, Native ARM compilation.	Used to build the mobile application's user interface and logic for both Android and iOS from a single codebase.

LiteRT (TensorFlow Lite)	Model quantization, Edge-AI inference, hardware acceleration support.	Serves as the runtime engine to execute the YOLOv8-Nano model locally on mobile hardware without internet.
YOLOv8-Nano	State of the art accuracy using an anchor-free detection head to predict pixel-level mask simultaneously	Instance segmentation, lightweight architecture, high inference speed.
Dart	Type-safe programming, efficient memory management.	The primary programming language used with Flutter to develop the application's functionality.
VS Code / Android Studio	Integrated Development Environment (IDE), debugging tools, emulator support.	Used as the main workspace for writing code, debugging the application, and managing the project files.
Roboflow	Polygon annotation tools, dataset versioning, and auto-augmentation.	Used to label the mangrove root dataset and export it in the YOLOv8 format for training.

Python	Extensive library ecosystem (PyTorch, Ultralytics), CUDA GPU acceleration, and automated training scripts.	Used as the primary environment for training the model, evaluating metrics, and converting the model to LiteRT format.

The software architecture is centered around the seamless integration of Roboflow, Python, Flutter, and LiteRT. Roboflow provides the foundation for data quality through precise polygon labeling for instance segmentation. This dataset is then fed into a Python-based environment, where the YOLOv8-Nano model is trained and enhanced using the Ultralytics framework and GPU acceleration. Once optimized, the model is converted into a LiteRT format. Flutter provides the front-end capability to interact with the device's camera and display real-time results, while LiteRT manages the heavy computational task of running the model. By using quantized neural networks, the software ensures that the application remains responsive even on mid-range Android devices, effectively managing memory and CPU threads to prevent hardware throttling during field assessments.

3.2.2 Hardware

The following hardware components are required to facilitate the development, training, and deployment of the system.

Table 2: *Hardware Requirements*

Hardware	Features	Purpose
Workstation/Laptop	Multi-core CPU, dedicated NVIDIA GPU (for training).	Used for training the YOLOv8 model, processing the dataset, and coding the Flutter application.
Android Smartphone	Camera (12MP+), 4GB+ RAM, NPU/GPU acceleration support.	The primary deployment device used by field researchers to capture images and run the MangroveGuard app.
USB Data Cable	High-speed data transfer.	Used for debugging the application directly on the mobile device during the development phase.

For development, a high-performance workstation is utilized to train the deep learning model, as training requires substantial GPU power to process labeled images. Once the model is optimized and converted to a LiteRT format, it is deployed onto a standard Android smartphone. The smartphone's internal camera serves as the sensor for data acquisition, while its mobile processor handles the inference, making the system a completely self-contained unit for remote fieldwork.

3.2.3 People-ware

The success of the MangroveGuard system depends on the interaction between various stakeholders and user types:

- **Developers/Researchers:** The capstone team responsible for data collection, model training, and application development. They ensure the technical integrity of the YOLOv8-Nano model and the stability of the Flutter UI.
- **Environmental Scientists:** Expert users who utilize the application to supplement their professional ecological surveys. They provide the ground-truth knowledge required to validate the application's assessment accuracy.
- **Local Stakeholders:** The primary field users who monitor protected mangrove areas. They use the application as an accessible tool to quickly identify areas of concern regarding mangrove stability without needing advanced technical training.
- **Local People**: The primary field users who monitor protected mangrove areas. They use the application as an accessible tool to identify areas of concern regarding mangrove stability without needing advanced technical training.

CHAPTER IV

This chapter presents the systematic approach used to develop the MangroveGuard system, covering the requirements analysis, architectural design, and implementation strategies. It details the transition from high-level conceptualization to the practical deployment of a deep learning model within a mobile environment.

Methodology, Results and Discussion

4.1 Requirements Analysis

Performance Requirement Analysis

Performance Requirements Analysis is an aspect of the requirement analysis that focuses on understanding, defining, and documenting any performance-related needs of the system. It is also documented the expectations for a system or software application. It involves the identifying and specifying the performance metrics, targets, and constraints the system must meet to ensure efficiency and effective operation of the created application.

The goal of Performance Requirements Analysis is to create objectives that the software part of the application must meet. It ensures that the system or software application runs optimally and functions the desired required performance. It aids in designing, developing, and testing efforts to address performance related issues and maximize the system's efficiency, responsiveness, and scalability.

In Performance Requirement Analysis, the following activities are typically carried out.

Table 3: *Performance Requirement Analysis*

Performance Requirements	Description	Acceptable Value
Model Performance	The system shall accurately detect and classify mangrove structural conditions using YOLOv8-Nano	At least 80% on validation dataset
Application Startup	The application shall load and become usable within a reasonable amount of time	≤ 5 seconds
Offline Functionality	The system shall perform detection without internet connection	100% offline inference capability
Inference Time	The system shall process a single captured image and display results within an acceptable response time	≤ 3 seconds per image on mid-range Android Device

Safety Requirement Analysis

Safety Requirement Analysis is an aspect of the Requirement Analysis that identifies and specifies the safety requirements that the application or system must

meet. It describes the risk and failures that may cause consequences to the human life environment and facilities that the application works on. It is essential that the system proposed is designed, developed, operated to minimize threat or risk and protect the people and the environment. Other than that, it also identifies any safety concerns that the system will face in the early development cycle.

In the Safety Requirement Analysis, the following information is the needed requirement for the safety of not just the system but also the other factors on where it is used.

Table 4: *Safety Requirement Analysis*

Safety Requirement	Description
Field Research Safety	Be able to provide warning message reminding the users to ensure safety in mangrove areas
AI Decision Safety	Be able to display prediction confidence scores and include a disclaimer that the results are assistive and not a replacement for expert ecological judgement

Security Requirement Analysis

Security Requirement Analysis is an aspect of Requirement Analysis that focuses on identifying, explaining and specifying the security of the system, application, or infrastructure. It ensures that the system or application is safe from malicious intent that would lead to exposure of data or modification. It is essential that the proposed system is designed, developed, and optimized to handle such threats that

the system would face. In the Security Requirement Analysis, the following information is needed for the security of the system.

Table 5: *Security Requirement Analysis*

Security Requirement	Description	Rationale
Activity Logging	The system shall log image assessment and processed information	Enables traceability and accountability for research documentation
Model File Integrity Protection	The model shall be protected from unauthorized modification	Prevents tampering of model which may result in inaccurate ecological as assessment

4.2 Design of Software, Systems, Product, and/or Processes

Systems Development Life Cycle (SDLC)

In this study, the researchers used the Iterative Waterfall Model. Founder of the model, Dr. Winston W. Royce, pioneered the notion in a paper published in 1970 about his experience designing satellite software. Dr. Royce did not use the term waterfall, instead he alluded to the downstream value of documentation. One method for developing software is the iterative waterfall approach which combines the sequential processes of the standard waterfall model with the flexibility of iterative design. Enhancement and adjustment can be made at each stage of the development process, rather than waiting until each part is finished until the end of the project. It includes feedback pathways from each step to the stages that came before the previous phase. If

issues surface during each stage, the feedback pathways allow the phase on which the mistake was stated, to be modified or fixed by the researchers or developers.

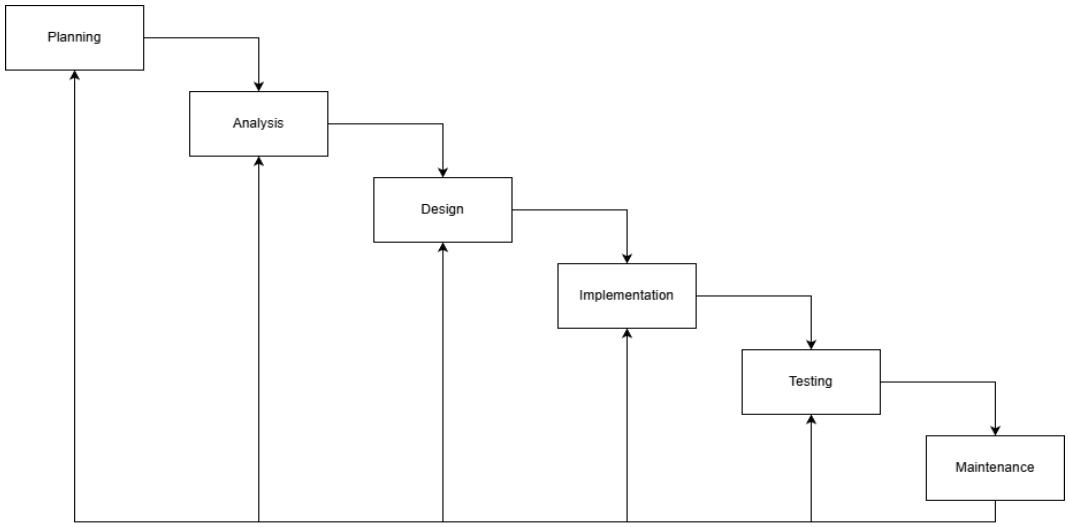

Figure 1: *Diagram Representation of the Iterative Waterfall Model*

Planning Phase

In this phase, the goal is to determine whether the application is technically, economically, operationally, and academically viable. For this project, the researchers assess whether YOLOv8-Nano can run efficiently on mobile devices, the datasets needed can be collected and the system can realistically analyze visible structural characteristics. The researchers also evaluate hardware and software requirements including mobile devices specifications, tools, training platforms, and deployment frameworks The plan in this phase includes conducting related literature review on deep learning in ecological monitoring, consulting with environmental agencies, estimating deployment time and verifying that the project can be completed within the capstone timeframe

4.2.2 GANNT Chart

Table 6: *GANTT CHART*

Code	List of Activities	Duration	Date (Start - Finish)
A1	Adviser Confirmation Letter Approval	1	01\19\2026-01\20\2026
B1	Approval of Concept Paper	1	01\20\2026-01\21\2026
C1	Chapter 1	15	01\19\2026-02\06\2026
D1	Approval of Interview Letter for DENR	1	01\21\2026-01\22\2026
E1	Approval of Letter for Visitation on Mangrove Sites	1	01\21\2026-01\22\2026
F1	Interview in DENR	1	01\22\2026-01\23\2026
G1	Collect Labled Datasets for Model on Selected Mangrove Sites	30	01\26\2026-03\06\2026
H1	Chapter 2	15	02\02\2026\-02\20\2026
I1	Training of Model for Application	30	02\23\2026\-04\10\2026
J1	Chapter 3	15	02\23\2026\-03\13\2026
K1	Creation of 50% of the System	30	02\16\2026\-03\27\2026
L1	Chapter 4	10	03\16\2026\-03\27\2026
M1	Compilation and Finalization of Paper	5	04\06\2026\-04\10\2026
N1	Preparation for Pre-Oral Defense	5	04\13\2026\-04\17\2026
O1	Pre-Oral Defense	5	04\20\2026\-04\24\2026
P1	Submission and Approval of Documents to REC for Clearnance	65	04\27\2026\-07\31\2026
Q1	Revision of Chapter 1 to 4 (Post-Defense)	25	04\27\2026\-05\29\2026
R1	Chapter 5	15	05\25\2026\-06\12\2026
S1	Creation of Appendices and References	15	06\08\2026-06\26\2026
T1	Testing of Model and Finalization of Mobile Application User Interfa	15	07\06\2026-07\24\2026
U1	Integration of Model to Mobile Application	15	07\20\2026-08\07\2026
V1	Overall Mobile Application Performance Evaluation	5	08\10\2026-08\14\2026
W1	ISO Evaluation	10	08\10\2026-08\21\2026
X1	Whitebox and Blackbox Testing	15	08\17\2026-09\21\2026
Y1	Refinement, Testing and Debugging	15	08\24\2026-09\11\2026
Z1	Installation of the System	5	09\14\2026-09\18\2026
A2	Preparation for Final Defense	15	09\14\2026-10\02\2026
B2	Final Defense	5	10\12\2026-10\16\2026
C2	Final Revision for Capstone Project Documents	25	10\12\2026-11\13\2026
D2	Secure Grammarian Certification	5	11\02\2026-11\06\2026
E2	Secure Plagiarism Report	5	11\09\2026-11\13\2026
F2	Submission of Capstone Project	1	11\16\2026-11\20\2026

Legend

On-Time

Critical Path

4.2.3 PERT Chart

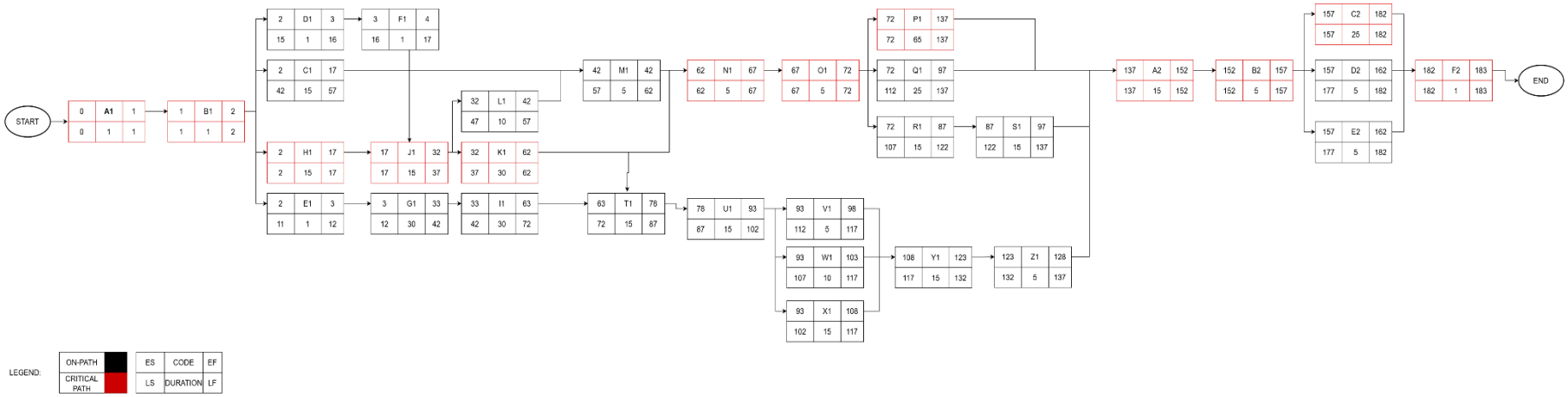

Figure 8: *PERT Chart Diagram*

4.2.4 Activity List

Table 7: *List of Activities*

Code	Activty Name	Duration
	Sprint 1	
A1	Adviser Confirmation Letter Approval	1
B1	Approval of Concept Paper	1
C1	Chapter 1	15
D1	Approval of Interview Letter for DENR	1
E1	Approval of Letter for Visitation on Mangrove Sites	1
F1	Interview in DENR	1
G1	Collect Labled Datasets for Model on Selected Mangrove Sites	30
H1	Chapter 2	10
H1.1	Related Literature	5
H1.2	Related System	5
I1	Training of Model for Application	30
	Sprint 2	
J1	Chapter 3	15
J1.1	Current System	5
J1.2	Proposed System	10
K1	Creation of 50% of the System	30
L1	Chapter 4	10
M1	Compilation and Finalization of Paper	5
N1	Preparation for Pre-Oral Defense	5
O1	Pre-Oral Defense	5
	Sprint 3	
P1	Submission and Approval of Documents to REC for Clearnance	65
Q1	Revision of Chapter 1 to 4 (Post-Defense)	25
R1	Chapter 5	15
S1	Creation of Appendices and References	15
T1	Testing of Model and Finalization of Mobile Application User Interface	15
U1	Integration of Model to Mobile Application	15
V1	Overall Mobile Application Performance Evaluation	5
W1	ISO Evaluation	10
	Sprint 4	
X1	Whitebox and Blackbox Testing	15
Y1	Refinement, Testing and Debugging	15
Z1	Installation of the System	5
A2	Preparation for Final Defense	15
B2	Final Defense	5
	Sprint 5	

C2	Final Revision for Capstone Project Documents	25
D2	Secure Grammarian Certification	5
E2	Secure Plagiarism Report	5
F2	Submission of Capstone Project	1

In *Table 6,* a Gantt chart of the system can be defined as a highly used tool in project management. In the bar chart, the tasks or activities mentioned are on a time scale, displaying the project schedule. It is a basic project-management mechanism, which provides a concise and brief description of each activity in terms of start and end date, duration, interrelationship, and overall project status done by the researchers.

On the other hand, Figure 7.1 to 7.3 represents the PERT Chart of the project. It is used to represent the task, milestone, dependencies, and general project timeline of the system. A PERT chart (acronym for Program Evaluation and Review Technique) is a graphical project-management tool that represents a project as a network diagram, therefore, depicting the task, milestones, dependencies, and the overall timeline.

Analysis Phase

During the requirements analysis phase, the focus is on defining system requirements and understanding user needs. For the **MangroveGuard App**, this includes identifying the target users as locals that wants to analyze the stability of the tree, and defining functional requirements such as guided image capture, on-device object detection of mangrove trunks and roots, and the generation of shareable PDF reports. Non-functional requirements are also specified, emphasizing high usability for in-field operation, performance efficiency on mobile devices, and complete offline capability, as all processing is done locally.

The researchers analyze which structural indicators are key for a mangrove stability scoring model, leading to the selection of six structural indicators Root Count

(RC), Root Density (RD), Root Spread (RS), Root Coverage Ratio (RCR), Structural Symmetry (SS), and Root Thickness Proxy (RT) as inputs to a weighted stability score. The plan includes creating requirement documents and use case diagrams, and defining system inputs (a camera image) and outputs (a stability classification and a detailed report). Actions performed include studying mangrove stability indicators, defining the evaluation metrics that result in the High, Moderate, and Low classifications, and structuring how the raw inference results from the TFLite model will be converted into a meaningful stability assessment using a fixed pixel-to-meter conversion.

Design Phase

In the design phase, the system architecture for the MangroveGuard App is structured based on the analyzed requirements. For this application, this includes designing the Flutter mobile application interface, configuring the TensorFlow Lite model architecture, and defining the stability scoring algorithm logic. The researchers design the user interface including the onboarding_page, scanner_page, and metrics_page to be simple and efficient for field use, with a guided camera frame to ensure effectiveness and satisfaction under ISO usability standards.

The on-device deep learning pipeline is also designed, detailing dataset preprocessing, annotation format for mangrove trunks and roots, model training parameters, and optimization for deployment as a .tflite file. This phase includes creating system flowcharts and wireframes, and formally defining the mangrove stability computation formula: $S = 0.20RC + 0.15RD + 0.20RS + 0.15RCR + 0.20SS + 0.10RT$, along with the fixed metersPerPixel conversion constant. Actions performed include dataset labeling, structuring the TFLite model training pipeline, defining the logic that connects the model's inference output to the stability score calculation, and

designing how the final classification (High, Moderate, Low) and scan history are displayed in the mobile app and exported to PDF reports.

4.2.5 Context Diagram

Figure 8: *Context Diagram of Current System*

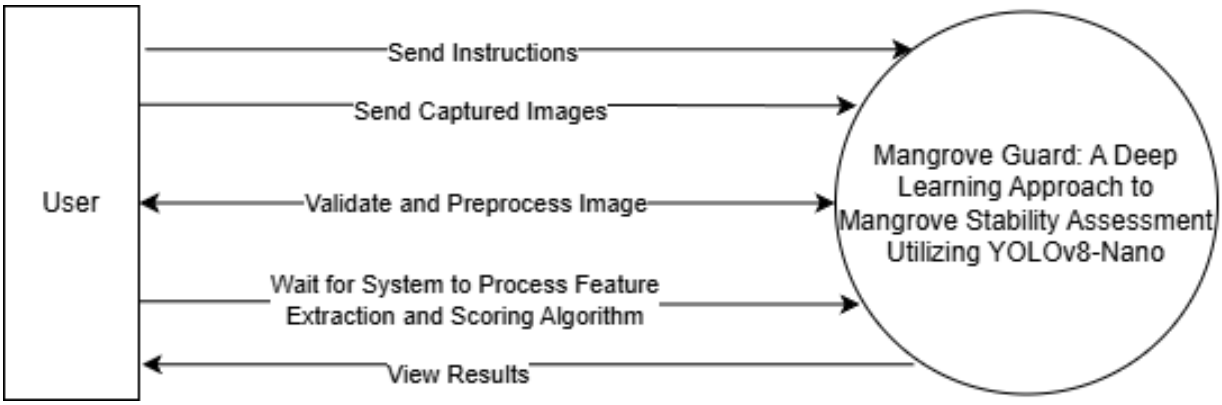

Figure 9: *Context Diagram of Proposed System*

4.2.6 Data Flow Diagram

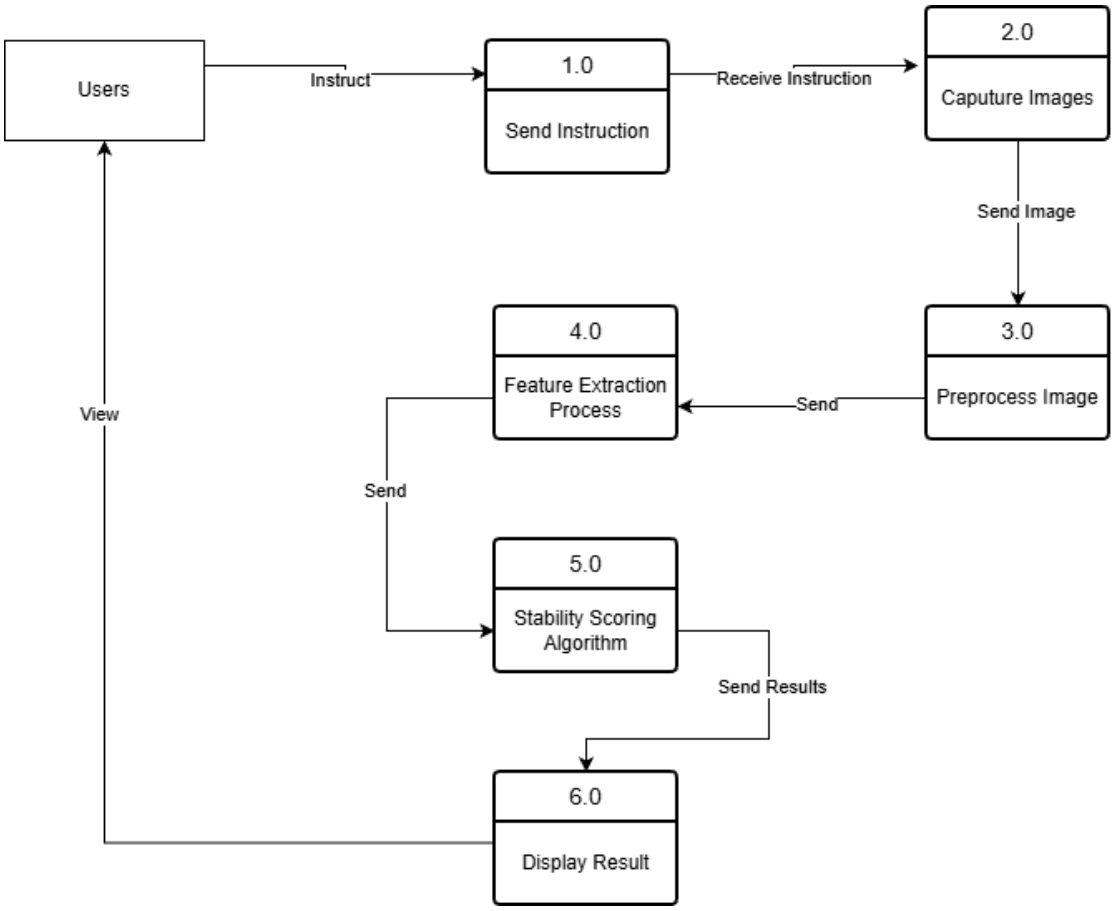

Figure 10: *Data Flow Diagram*

4.2.7 System Flowchart

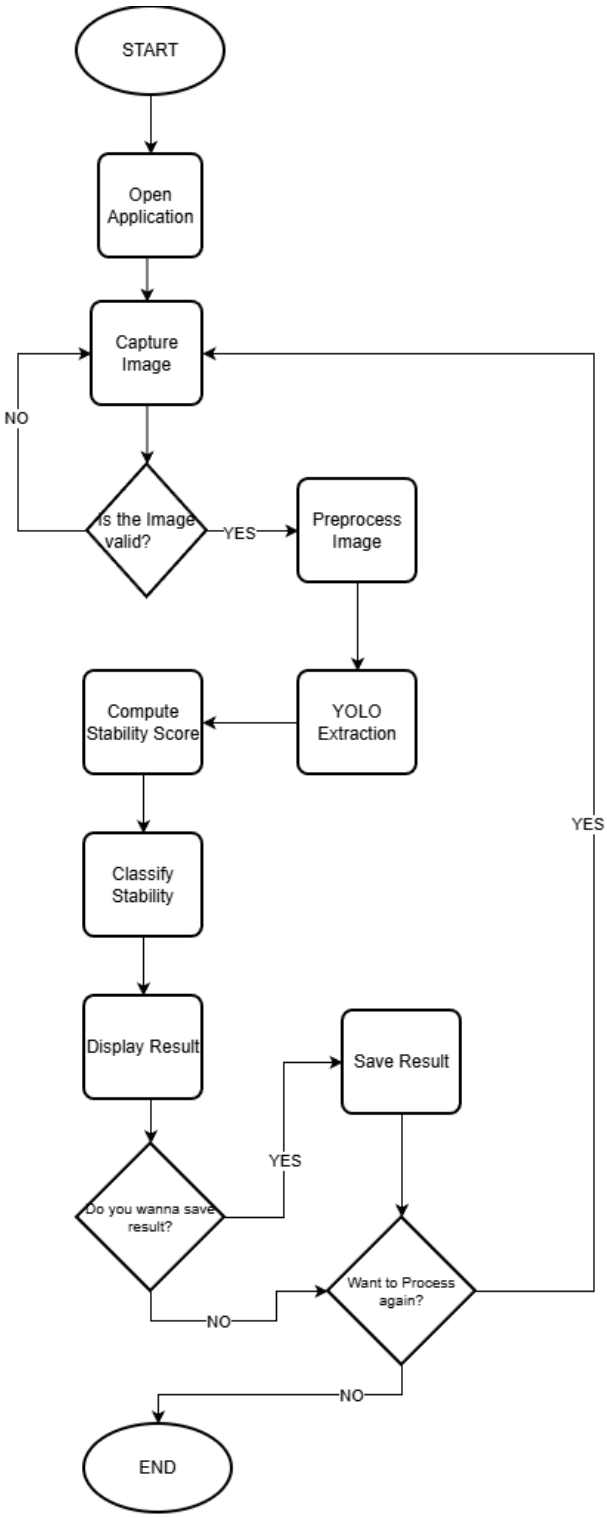

Figure 11: *System Flowchart*

4.2.8 Use Case Diagram

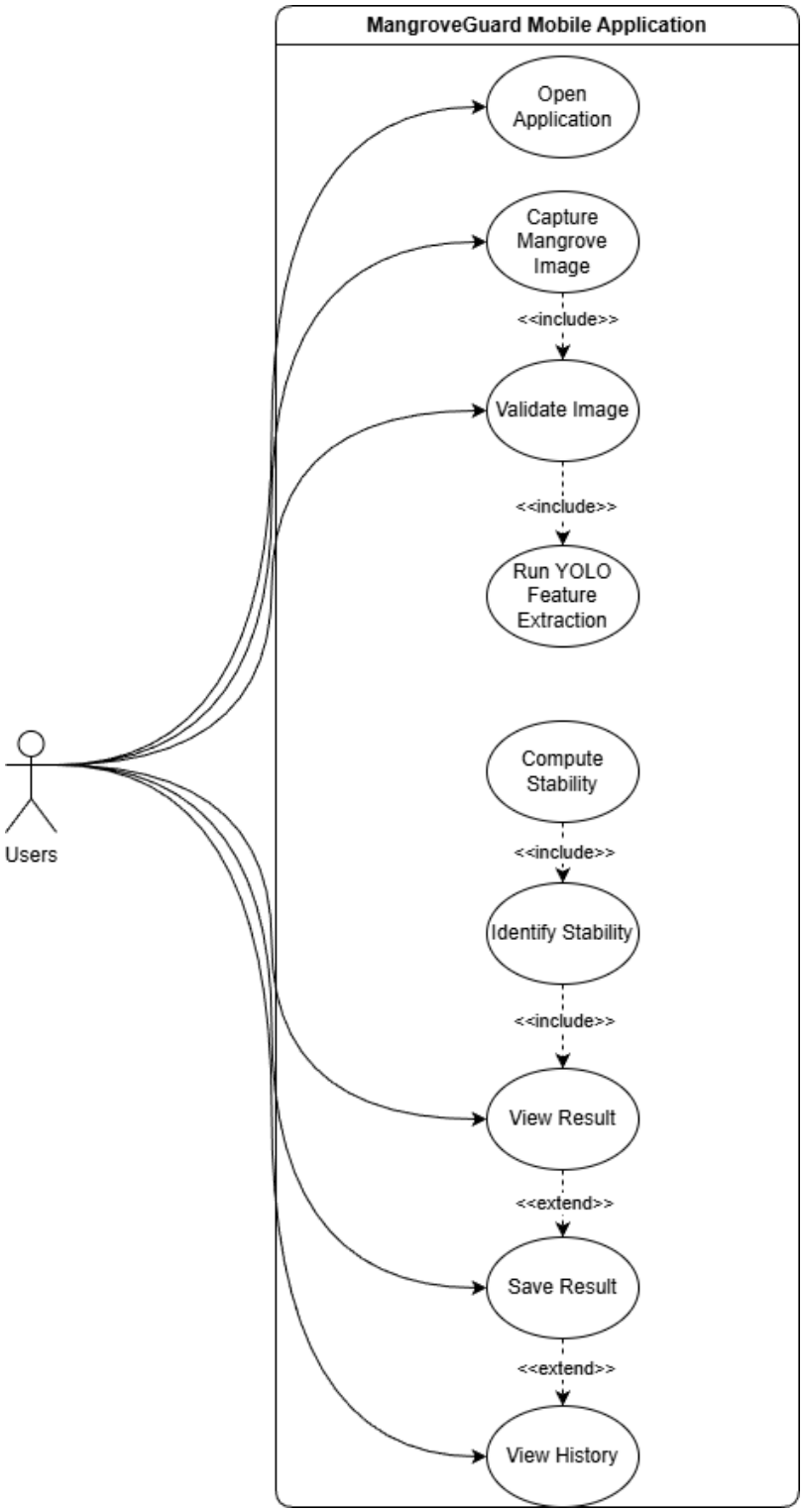

Figure 12: *Use Case Diagram*

Implementation Phase

In the implementation phase, the actual development of the Mangrove Guard App takes place. For this application, this involves training a computer vision model on annotated mangrove images to identify key features like trunk and roots, developing the Flutter mobile application, integrating the trained model into the app, and implementing the stability scoring algorithm. The developers write the source code using Dart and Flutter for the main application modules, including the onboarding flow, the core scanner_page.dart for image capture, and pages for viewing metrics and recent scans.

The camera package is used to get a live feed from the device's camera, and the tflite_flutter package is used to load the mangroveModel.tflite file and perform real-time, on-device inference. A key action is implementing the stability score computation. This logic processes the bounding box outputs from the model, converts detected root and trunk measurements into normalized features using a fixed metersPerPixel constant, then computes RC, RD, RS, RCR, SS, and RT. These indicators are combined into a weighted stability score $S = 0.20RC + 0.15RD + 0.20RS + 0.15RCR + 0.20SS + 0.10RT$, which is used to classify stability as High, Moderate, or Low.

The plan includes iterative model training to improve detection accuracy, exporting the final optimized .tflite model file for mobile deployment, and ensuring the detection outputs are correctly piped into the stability computation logic. Further implementation details include using the shared_preferences package to persist the onboarding state and recent scan history, and leveraging the pdf package to generate reports. For Android, a specific native method channel, mangroveguardapp/downloads,

is implemented to handle saving the exported PDF reports directly to the device's "Downloads" folder. The entire process is designed for on-device execution, requiring no external backend for its core analysis features.

Testing Phase

The testing phase ensures that the **Mangrove Guard App** functions correctly, delivers accurate stability assessments, and meets performance expectations for field use. For this application, testing includes evaluating the on-device TensorFlow Lite model's ability to accurately extract RC, RD, RS, RCR, SS, and RT, which are the inputs to the weighted stability score used by the core stability logic. Crucially, this phase involves usability testing based on ISO 9241-11 standards (effectiveness, efficiency, and satisfaction), focusing on the guided camera workflow and the clarity of the final stability classification (High, Moderate, Low).

Researchers also conduct extensive testing in real-world field scenarios to evaluate the system's robustness against variable lighting, water reflections, and interference from dense, overlapping root structures. The plan includes functional testing of all app features (camera, inference, PDF export), performance testing to monitor battery consumption and UI responsiveness during live scanning, and validation of the app's stability scores against assessments from expert ecologists. Actions performed include analyzing the accuracy of the calculated stability score, conducting user testing sessions with field researchers to gather feedback on the app's workflow and the utility of the generated PDF reports, identifying common detection or measurement errors, and refining the model or the metersPerPixel constant accordingly. This feedback loop is critical and may require iterating back to the implementation phase to improve the system's reliability and practical value.

Maintenance Phase

After deployment, the maintenance phase focuses on the long-term improvement and reliability of the MangroveGuard App based on field feedback and environmental variability. For this application, this includes periodically retraining the mangroveModel.tflite with new mangrove images to improve its generalization across different species, lighting conditions, and geographical locations. It also involves fixing bugs in the Flutter application, such as issues with the camera plugin on certain devices, PDF export failures handled by the native Android mangroveguardapp/downloads channel, or data persistence problems with shared_preferences.

The plan includes periodic dataset expansion, model re-training, and publishing app updates to the respective app stores. A key maintenance activity is refining the business logic, such as adjusting the fixed metersPerPixel conversion factor or the stability classification thresholds if field data suggests a more accurate model is needed. Actions performed involve monitoring the app's stability score accuracy against expert assessments, addressing user-reported issues, updating dependencies in the pubspec.yaml file to ensure security and compatibility, and refining the UI on pages like scanner_page.dart to improve field efficiency. Since mangrove ecosystems are dynamic, continuous dataset updates and model fine-tuning are expected as part of the app's long-term lifecycle to ensure it remains a valuable tool for researchers.

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